

NE9 Coursework 3: Advanced Fission and Fusion

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1 Pebble-bed modular reactor (PBMR)

(a) Why is a high core outlet temperature desirable for achieving Generation IV reactor goals?

A high core outlet temperature means a higher thermal efficiency. This promotes **Sustainability**, since more electricity is produced for less fuel. In theory, this is also good for **Economics**, because more electricity can be sold; this is obviously dependent on the price of the reactor itself.

The higher temperature also enables *process heat* to be used for decarbonizing industrial processes (especially hydrogen production). This is in line with the Gen IV objective of **Sustainability**, but also **economics** because such additional applications may generate more profits.

Because this exercise is about PBMRs, I feel the need to mention that high outlet temperatures are typically achieved with a gas coolant. This can be a boon to safety because of the lower power density, pressure, and lack of voiding. However, this benefit is not inherent to a high outlet temperature, so I'm not sure if this is relevant to the question.

(b) Graphite-moderated gas-cooled pebble-bed reactors are relatively neutronically inefficient. They require higher enrichment than Light Water Reactors and have lower fuel utilisation. Explain the reasons for these inefficiencies.

First off, carbon atoms are heavy, so graphite is a **worse moderator** than water. The harder spectrum means more resonance captures compared to fissions, so a worse neutron economy.

Furthermore, the fuel in the pebbles has a high surface-to-volume ratio. This means that there is **less U-238 self shielding**. Normally, a large portion of the absorption peaks are captured by the outer U-238, such that the inner U-238 captures fewer neutrons. This doesn't happen as much when the neutrons have less volume to travel through, therefore more neutrons get absorbed.

Lastly, pebble bed reactors are high-temperature reactors. These reactors need a **taller core** to allow for more decay heat removal. This increases neutron leakage.

(c) A PBMR is designed with the properties provided in the following table. Assume that the core power is distributed as a cosine in the axial direction: $q'(z) = q_0 \cos(\pi z/H)$ where H is the core height. Assume that the radial power distribution is flat. Calculate the surface temperature of the pebbles at a distance of 12 m downstream of coolant entry into the core.

First, find the temperature of the coolant. Since the power is proportional to $\cos(\pi z/H)$, the fraction of heat transferred to the coolant at axial distance z is:

$$\frac{P(z)}{P_{\text{tot}}} = \frac{\int_{-H/2}^z \cos(\pi z/H) dz}{\int_{-H/2}^{H/2} \cos(\pi z/H) dz}, \quad (1)$$

where $P(z)$ is the power transferred and P_{tot} is total core power. Solve and simplify.

$$\begin{aligned} \frac{P(z)}{P_{\text{tot}}} &= \frac{\sin(\pi z/H) \big|_{-H/2}^z}{\sin(\pi z/H) \big|_{-H/2}^{H/2}} \\ &= \frac{\sin(\pi z/H) + 1}{2} \end{aligned} \quad (2)$$

In this case, $z = 3$ and $H = 18$, which results in:

$$\frac{P(z)}{P_{\text{tot}}} = \frac{3}{4}. \quad (3)$$

Since 3/4 of the heat has been added, the coolant temperature becomes:

$$\begin{aligned} T_{\text{He}}(z = 3 \text{ m}) &= 500 \text{ }^{\circ}\text{C} + \frac{3}{4}(900 \text{ }^{\circ}\text{C} - 500 \text{ }^{\circ}\text{C}) \\ &= 800 \text{ }^{\circ}\text{C} \end{aligned} \quad (4)$$

Now, find the volumetric heat generation:

$$P_{\text{tot}} = \iiint q''' dV. \quad (5)$$

The radial distribution is flat, and the axial distribution is given. Therefore,

$$P_{\text{tot}} = \pi R^2 q_0 \int_{-H/2}^{H/2} \cos(\pi z/H) dz, \quad (6)$$

where R is the radius of the core. Solve the integral and simplify:

$$\begin{aligned} P_{\text{tot}} &= \pi R^2 q_0 \left. \frac{H}{\pi} \sin(\pi z/H) \right|_{-H/2}^{H/2} \\ &= 2HR^2 q_0. \end{aligned} \quad (7)$$

Solve for q_0 .

$$q_0 = \frac{P_{\text{tot}}}{2HR^2}. \quad (8)$$

Substitute (8) into $q'''(z) = \cos(\pi z/H)$.

$$q'''(z) = \frac{P_{\text{tot}} \cos(\pi z/H)}{2HR^2}. \quad (9)$$

So the local heat generation at $z = 3 \text{ cm}$ is

$$q'''(z = 3 \text{ cm}) = 6.415 \text{ MW/m}^3. \quad (10)$$

The average density of the pebbles is

$$\frac{\text{pebbles}}{H\pi R^2} = 5305.16 \text{ m}^{-3}. \quad (11)$$

Thus, the power per pebble P_p at $z = 3 \text{ cm}$ is:

$$\begin{aligned} P_p &= \frac{6.415 \text{ MW}}{5305.16} \\ &= 1209.20 \text{ W}. \end{aligned} \quad (12)$$

Now, the temperature difference can be found using the heat transfer coefficient h :

$$\Delta T = \frac{P_p}{hA}, \quad (13)$$

where A is the area of a pebble. Since the pebble is a sphere, the result becomes:

$$\begin{aligned} \Delta T &= \frac{P_p}{h4\pi r^2} \\ &= 106.92 \text{ }^{\circ}\text{C}. \end{aligned} \quad (14)$$

The coolant temperature was given in (4). Therefore, the surface temperature of the pebbles at $z = 3 \text{ cm}$ is:

$$800 \text{ }^{\circ}\text{C} + 106.92 \text{ }^{\circ}\text{C} = \boxed{906.92 \text{ }^{\circ}\text{C}}. \quad (15)$$

2 D-D fusion

For the sake of clarity, I will list all of the relevant reactions for the entire exercise here.

- First D-D reaction:



- Second D-D reaction:



- D-D full burn:



- D-T fusion:



- (a) Calculate the average energy released per deuteron corresponding to the complete burn, and compare that figure to the energy released per deuteron in the basic D-D reactions.

Start with the first reaction shown in (16). The Q value is given by:

$$\begin{aligned} Q/c^2 &= 2M({}_1^2\text{D}) - M({}_2^3\text{He}) - m({}_0^1\text{n}) \\ &= 2 \cdot 2.0141 \text{ u} - 3.01603 \text{ u} - 1.0087 \text{ u} \\ &= 0.00347 \text{ u} , \end{aligned} \quad (20)$$

so the energy per deuteron is:

$$\begin{aligned} \Delta E/\text{deu.} &= \frac{1}{2} 0.00347 \cdot 931.494 \text{ MeV} \\ &= \boxed{1.616 \text{ MeV}} . \end{aligned} \quad (21)$$

For the second reaction given in (17), do the same procedure.

$$\begin{aligned} Q/c^2 &= 2M({}_1^2\text{D}) - M({}_1^3\text{T}) - M({}_1^1\text{H}) \\ &= 2 \cdot 2.0141 \text{ u} - 3.01605 \text{ u} - 1.0079 \text{ u} \\ &= 0.00425 \text{ u} , \end{aligned} \quad (22)$$

$$\begin{aligned} \Delta E/\text{deu.} &= \frac{1}{2} 0.00425 \cdot 931.494 \text{ MeV} \\ &= \boxed{1.979 \text{ MeV}} . \end{aligned} \quad (23)$$

The full burn is given in (18). It is derived by combining (16) and (17) with 2 more deuterons, such that both the triton and helium nuclei react. Find the energy per nuclide:

$$\begin{aligned} Q/c^2 &= 6M({}_1^2\text{D}) - 2M({}_2^4\alpha) - 2M({}_1^1\text{H}) - 2m({}_0^1\text{n}) \\ &= 6 \cdot 2.0141 \text{ u} - 2 \cdot 4.0026 \text{ u} - 2 \cdot 1.0079 \text{ u} - 2 \cdot 1.0087 \text{ u} \\ &= 0.0462 \text{ u} , \end{aligned} \quad (24)$$

$$\begin{aligned} \Delta E/\text{deu.} &= \frac{1}{6} 0.0462 \cdot 931.494 \text{ MeV} \\ &= \boxed{7.173 \text{ MeV}} . \end{aligned} \quad (25)$$

Comparing the result in (25) to the ones in (21) and (23), it is evident that the complete burn releases a lot more energy per deuteron than the basic D-D reactions.

- (b) One advantage of the fully-catalysed D-D reaction compared to a D-T fusion reaction is that a larger fraction of the energy released is delivered to charged particles (H, T, He3, He4, ...), which can be contained by magnetic fields.
- i. Calculate the fraction of energy in charged particles for complete burn, and compare this to the value for the D-T reaction (...). Why is having a large fraction of energy delivered to the charged particles preferable?

Consider a reaction that produces 2 particles. From the center of mass frame, the momentum of both particles must cancel to 0, such that $|p_1| = |p_2|$. Let p denote the absolute momentum of both particles. The kinetic energy E of both particles is given by $E = p^2/(2m_i)$, where m_i is the mass of the particle in question. Since the momentum is constant, it is evident that $E \propto 1/m$. The following relation is derived:

$$E_i/E_{\text{tot}} = \frac{1/m_i}{1/m_1 + 1/m_2} . \quad (26)$$

The relationship in (26) will be used to determine the energy fractions.

Starting with the D-T reaction shown in (19), the only charged particle is the alpha. Thus, the fraction of energy delivered to charged particles is given by:

$$\begin{aligned} E_\alpha/E_{\text{tot}} &= \frac{1/M({}_2^4\alpha)}{1/M({}_2^4\alpha) + m({}_0^1\text{n})} \\ &= \frac{1/4.0026}{1/4.0026 + 1/1.0087} \\ &= \boxed{20.13\%} . \end{aligned} \quad (27)$$

Now move on to D-D fusion. It is easiest to go reaction by reaction, and calculate the fraction that goes to the *neutral* particles first. Starting with the reaction in (16), the energy fraction of the neutron is:

$$\begin{aligned} E_n/E_{\text{tot},1} &= \frac{1/m({}_0^1\text{n})}{1/M({}_2^3\text{He}) + m({}_0^1\text{n})} \\ &= \frac{1/1.0087}{1/3.01603 + 1/1.0087} \\ &= 0.749374. \end{aligned} \quad (28)$$

But this only considers the partial reaction (16). The contribution of this partial reaction to the total reaction is found by dividing the energy of the partial reaction in (20) by that of the total in (24):

$$\begin{aligned} E_{n,1}/E_{\text{tot}} &= 0.749374 \frac{0.00347}{0.0462} \\ &= 0.0562842. \end{aligned} \quad (29)$$

The same procedure is repeated for the second partial reaction. In this case, the only neutron produced occurs in the reaction shown in (19). The energy fraction going to the neutron is calculated in the same way as (27), except that the neutron is considered:

$$E_n/E_{\text{tot},2} = 0.798715. \quad (30)$$

The Q value of this equation is:

$$\begin{aligned} Q/c^2 &= M({}_1^2\text{D}) + M({}_1^3\text{T}) - M({}_2^4\text{He}) - m({}_0^1\text{n}) \\ &= 2.0141 \text{ u} + 3.01605 \text{ u} - 4.0026 \text{ u} - 1.0087 \text{ u} \\ &= 0.01885 \text{ u}. \end{aligned} \quad (31)$$

Therefore, the energy fraction of the second neutron in (30) is found by dividing (31) by (24):

$$\begin{aligned} E_{n,2}/E_{\text{tot}} &= 0.798715 \frac{0.01885}{0.0462} \\ &= 0.325883. \end{aligned} \quad (32)$$

The energy fraction for each of the neutrons is now known in and given in (29) and (32). Using these, the fraction going to charged particles can be found:

$$\begin{aligned} E_{\text{charged}}/E_{\text{tot}} &= 1 - 0.0562842 - 0.325883 \\ &= \boxed{61.78\%}. \end{aligned} \quad (33)$$

Clearly, the fraction going to charged particles in the D-D reaction (33) is much greater than for the D-T reaction (27). The most important benefit of this is that charged particles are sensible to the EM field. This means that they tend to stay near the plasma. Therefore, a large portion of their energy will be deposited directly into the plasma. This means that the plasma has an easier time sustaining itself.

Other benefits are that they have a predictable range (Bragg curve), they are easier to shield, and cause less structural and radioactive damage.

ii. Calculate the same fraction of energy in charged particles, if the tritium is completely burned but the He3 escapes before interacting with the D. Explain briefly why this might be likely to occur.

The total reaction for this exercise consists of the partial reactions shown in (16), (17), and (19). In this case, neutral particles are created in the first D-D reaction (16), as well as the D-T fusion (19). For both partial reactions, the fraction of energy going to the neutrons was already determined in (28) and (30). Specifically, for (16) that fraction is 0.749374, and for (19) that fraction is 0.798715.

To find the effect of these fractional reactions to the total reaction, it is necessary to find the Q value of the total reaction. The Q values for the partial reactions are given in (20), (22), and (31) respectively. These are: 0.00347 u for (16), 0.00425 u for (17), and 0.01885 u for (19). The Q value for the full reaction becomes:

$$\begin{aligned} Q/c^2 &= 0.00347 \text{ u} + 0.00425 \text{ u} + 0.01885 \text{ u} \\ &= 0.02657 \text{ u}. \end{aligned} \quad (34)$$

Therefore, the energy fraction of the neutron in (16) is:

$$\begin{aligned} E_{n,1}/E_{\text{tot}} &= 0.749374 \frac{0.00347}{0.02657} \\ &= 0.097867. \end{aligned} \quad (35)$$

The energy fraction of the neutron in (19) is:

$$\begin{aligned} E_{n,1}/E_{\text{tot}} &= 0.798715 \frac{0.01885}{0.02657} \\ &= 0.566646. \end{aligned} \quad (36)$$

From (35) and (36), the fraction going to charged particles is determined:

$$\begin{aligned} E_{\text{charged}}/E_{\text{tot}} &= 1 - 0.097867 - 0.566646 \\ &= \boxed{33.55\%}. \end{aligned} \quad (37)$$

Comparing (37) to (33), it becomes clear that the fraction of energy going to charged particles is much lower in this case.

The reason why ${}^3\text{He}$ leaks out, is for the simple reason that D-He3 fusion just has a lower cross-section at the relevant (lower) temperatures. Both D-D, and especially D-T, have much higher cross-sections so deuterons are much more likely to interact with them. This means that ${}^3\text{He}$ is unlikely to fuse before it escapes the plasma.

3 D-T Fusion and Tritium breeding

- (a) *All deuterium-tritium fusion reactors will require a breeder blanket structure around the reaction chamber. Briefly outline the role of this piece of technology and a few of its vital features.*

The purpose of these blankets is to absorb neutrons. This has 3 main functions: they **breed tritium** for the tritium-consuming D-T fusion, they **convert power** from the neutrons into heat to use for electricity generation, and they **shield** the outside from the radioactive neutrons. The blanket has to capture radioactive particles, so it needs to withstand the radiation from the neutrons and the charged particles. Since the blanket sits inside the torus, it also needs to be resistant to high temperatures and EM fields.

- (b) *For a breeder-blanket, approximately 20% of a fusion reactor interior will be taken up with the divertor, access ports and other structures and is therefore unavailable for blanket placement.*
- i. *Estimate the breeder-blanket minimum neutron multiplication factor (neutrons produced/neutrons absorbed) required for a continuous fusion reaction, if 98% of the neutrons are retained in the blanket and 90% of these are captured by the breeder.*

Each neutron must produce at least 1 more neutron for a continuous fusion reaction. It is given that around 80% of the neutrons enter the blanket, 98% of these are retained, and 90% interact with the breeder. These are then multiplied by the neutron multiplication factor (NMF), so:

$$0.80 \cdot 0.98 \cdot 0.90 \cdot \text{NMF} = 1. \quad (38)$$

Solve for NMF:

$$\text{NMF} = 1.42. \quad (39)$$

- ii. *Calculate the blanket thickness required in a D-T fusion reactor...*

It was given in the previous question that 2% of the neutrons make it through the blanket. Therefore:

$$e^{-t/\lambda} = 0.02, \quad (40)$$

where $\lambda = 15$ cm (given). Solving for t gives:

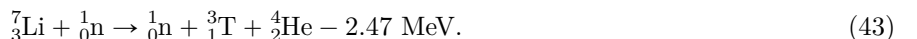
$$t = \boxed{58.7 \text{ cm}}. \quad (41)$$

- (c) *The fuel must be bred within the reactor using a lithium blanket. Lithium has two stable isotopes Li6 and Li7. How do the neutron-induced reactions that generate tritium differ between these nuclides? Describe a spatially heterogeneous enrichment strategy that would maximise tritium production in a fusion reactor blanket. What else could be added to increase the tritium production rate?*

The reaction for ${}^6\text{Li}$ is



while the reaction for ${}^7\text{Li}$ is



So there are a couple of important differences: The reaction for ${}^6\text{Li}$ is **exothermic**, while the reaction for ${}^7\text{Li}$ is **endothermic**. As a consequence, ${}^6\text{Li}$ has a **much higher cross-section at low neutron energies** than that

of ${}^7_3\text{Li}$. Additionally, the reaction for ${}^7_3\text{Li}$ **does not consume a neutron**. This is important, because the extra neutron can breed another ${}^3_1\text{T}$, thus increasing the tritium breeding ratio.

From these observations, it can be hypothesized that a spatially heterogeneous blanket may be beneficial to maximize tritium production. Specifically, the inner blanket (facing the plasma) should be made from ${}^7_3\text{Li}$, which should gradually transition to ${}^6_3\text{Li}$ for the outer blanket (away from the plasma). This is because the neutrons entering the blanket are fast, so there is no cross-section penalty for using ${}^7_3\text{Li}$, and the extra neutron production can maximize tritium breeding. But as the neutrons reach the other side, they are slowed down due to scatter or interactions with ${}^7_3\text{Li}$. Now, it is more important to maximize the cross-section, so ${}^6_3\text{Li}$ becomes a favorable.

Tritium production can be further increased by adding **neutron multipliers**. These are elements with a high (n , $2n$) cross-section and low chance of absorption. Such multipliers increase the number of neutrons, who will then increase the amount of tritium production. Beryllium and lead are common examples of such elements.

- (d) *If breeding blankets were not in place, tritium would need to be source from elsewhere. If every D-T fusion reaction releases 17.6 MeV and fusion occurs exclusively through D-T fusion, what is the annual tritium fuelling requirement for a 2 GWth fusion plant (assuming no losses or multiplication effects)? At a cost of £50,000 per gram, what would be the cost?*

The thermal power of the fusion plant P is equivalent to:

$$\begin{aligned} P &= 2 \text{ GJ/s} \\ &\equiv 2 \cdot 10^9 \cdot 6.241509 \cdot 10^{12} \text{ MeV/s} \\ &= 1.248301 \cdot 10^{22} \text{ MeV/s}. \end{aligned} \tag{44}$$

Since each reaction produces 17.6 MeV, that implies a total reaction rate of:

$$\frac{1.248301 \cdot 10^{22} \text{ MeV/s}}{17.6 \text{ MeV}} = 7.092619 \cdot 10^{20} \text{ s}^{-1}. \tag{45}$$

The amount of tritium consumed per second is therefore:

$$\frac{7.092619 \cdot 10^{20} \cdot M({}^3_1\text{T})}{N_A}, \tag{46}$$

where N_A is the Avogadro constant. Assuming that the reactor runs for every second in a year, and that each year has 365.25 days, the amount of tritium needed each year is:

$$\begin{aligned} 365.25 \cdot 86400 \frac{7.092619 \cdot 10^{20} \cdot M({}^3_1\text{T})}{N_A} \text{ g} &= 365.25 \cdot 86400 \frac{7.092619 \cdot 10^{20} \cdot 3.01605}{6.02214076 \cdot 10^{23}} \text{ g} \\ &= \boxed{112 \text{ kg}}. \end{aligned} \tag{47}$$

Given an annual cost of £50000 per g, the total annual cost becomes:

$$£50000 \cdot 112000 = \boxed{£5.6 \text{ Billion}}. \tag{48}$$